A STANDARD PH COVERING IS A STANDARD V COVERING

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [3658, 3661] and proof(s) [[3663, 3693]]; this spec covers the complete environment instead (statement 3658-3661, proof span(s) [(3663, 3693)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original definition 0DBD: sources/stacks-chow/tags.txt, source lines 16693–16693. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00IA: sources/stacks-chow/tags.txt, source lines 672–672. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0BX5: sources/stacks-chow/tags.txt, source lines 14968–14968. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *A standard ph covering is a standard V covering.*

Proof. Let T be an affine scheme. Let $f : U \to T$ be a proper surjective morphism. Let $U = \bigcup_{j=1,\ldots,m} U_j$ be a finite affine open covering. We have to show that $\{U_j \to T\}$ is a standard V covering, see Definition 0DBD. Let $g : \mathrm{Spec}(V) \to T$ be a morphism where V is a valuation ring with fraction field K. Since $U \to T$ is surjective, we may choose a field extension L/K and a commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(L) & & \longrightarrow & & U \\ \downarrow & & & & \downarrow \\ \mathrm{Spec}(K) & \longrightarrow & \mathrm{Spec}(V) & \xrightarrow{g} & T \end{array}$$

By Algebra, Lemma 00IA we can choose a valuation ring $W \subset L$ dominating V. By the valuative criterion of properness (Morphisms, Lemma 0BX5) we can then find the morphism h in the commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(L) & \longrightarrow & \mathrm{Spec}(W) & \xrightarrow{h} & U \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spec}(K) & \longrightarrow & \mathrm{Spec}(V) & \xrightarrow{g} & X \end{array}$$

Since $\mathrm{Spec}(W)$ has a unique closed point, we see that $\mathrm{Im}(h)$ is contained in U_j for some j. Thus $h : \mathrm{Spec}(W) \to U_j$ is the desired lift and we conclude $\{U_j \to T\}$ is a standard V covering. □

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